Study Guide MD-102 — Endpoint Administrator

🌐 **Language:** English | Nederlands

Duration: 7 weeks · **Lab preset:** Standard (DC01 · MGMT01 · W11-01 · W11-02) + W11-AUTOPILOT for week 5 **MS Learn path:** Endpoint Administrator **Exam weight:**

Domain	Weight
Prepare infrastructure for devices	25–30%
Manage and maintain devices	30–35%
Manage applications	15–20%
Protect devices	15–20%

> **Updated:** Skills measured as of January 23, 2026. The domain structure changed significantly — "Deploy and upgrade" and "Manage identity and compliance" merged into **Prepare infrastructure for devices**. "Manage, maintain, and protect" was split into two separate domains.

> **Prerequisite:** MSDN/Visual Studio subscription with Microsoft 365 E5/E3 developer tenant (for Intune, Entra ID) **Tenant note:** when an exercise mentions Intune, Entra, or portal steps, perform those in the shared Sogeti / Capgemini dev tenant and use the local VMs as your test devices. **Domain note:** your own verified custom domain in the dev tenant is not required, but it is useful when you want more realistic UPN, Entra Connect, and hybrid identity practice.

How to use this study guide

- Start each week by reading the learning objectives and MS Learn modules so you know which exam scope you are covering before opening the lab.
- Then perform the lab exercises on the listed VMs and write down what you observe in the portal, PowerShell, and on the client devices.
- Complete the knowledge check only after finishing both the theory and the lab; use the answers to identify weak spots in understanding, not just to verify memorisation.
- If you work with both Dutch-speaking and English-speaking colleagues, deliberately keep both terms in mind, for example *compliance policy / compliancebeleid* and *hybrid join / hybride join*.

Lab coverage and expectations

- **Strong SSW-Lab coverage:** Windows deployment, Intune enrollment, Autopilot fundamentals, configuration profiles, compliance, update management, app deployment, endpoint security, and LAPS.
- **Partial coverage:** co-management, Windows 365, Endpoint Privilege Management, advanced reporting, and some Intune Suite features require extra tenant capabilities or remain partly conceptual in this lab.
- **Not fully reproducible locally:** fast-moving Microsoft UI changes, licence-dependent features, and features only visible in a fully provisioned Microsoft 365 tenant.
- Use this guide as a complete study path, but be honest about the gaps: if a topic feels only partially covered, revisit the matching MS Learn module directly in the portal.

How to use the knowledge checks

- Answer every question without looking at the solution first.
- Explain your answer out loud as if you are handing over the topic to a colleague.
- Revisit only the questions you missed or answered with uncertainty.
- Pay special attention to scenario questions: the exam often tests whether you can choose the right approach, not just recognise the term.

Week 1 — Windows client deployment

> **Exam domain:** Prepare infrastructure for devices · **Weight:** 25–30%

> **Real-world scenario:** A consultant at a mid-sized financial services firm is tasked with rolling out Windows 11 to 400 desktops currently running Windows 10 21H2. The IT manager wants to preserve existing applications and user profiles but also needs a clean deployment method for 50 new machines arriving from the OEM. The consultant must choose between in-place upgrade and wipe-and-load, prepare answer files for the new hardware, and validate that all devices meet the Windows 11 hardware requirements before the project starts.

Learning Objectives

- ☐ Identify the minimum hardware requirements for Windows 11 (TPM 2.0, UEFI + Secure Boot, 64 GB storage, 4 GB RAM)
- ☐ Distinguish between wipe-and-load, in-place upgrade, and fresh-start deployment methods and select the right method for a given scenario
- ☐ Explain the role of `autounattend.xml` and how Windows System Image Manager (SIM) is used to create answer files
- ☐ Describe what DISM does versus what Sysprep does, and when each tool is appropriate

- ☐ Explain the purpose of `oscdimg.exe` in creating bootable ISO images for unattended deployments
- ☐ Identify the minimum Windows 11 build number required for Intune MDM enrollment

MS Learn modules

- Deploy Windows 11
- Upgrade Windows client
- Windows Deployment Services and imaging

Key Concepts

Term	Description
Wipe-and-load	Replaces the existing OS entirely with a fresh Windows installation; all user data and applications must be migrated separately
In-place upgrade	Upgrades Windows while preserving existing apps, settings, and user data; performed using Windows Setup (`setup.exe`)
autounattend.xml	An XML answer file placed on installation media that automates Windows Setup responses (language, product key, disk partitioning, etc.)
Windows SIM	Windows System Image Manager — the GUI tool for authoring and validating `autounattend.xml` answer files; included with the Windows ADK
DISM	Deployment Image Servicing and Management — a command-line tool for mounting, inspecting, and modifying Windows images (`.wim` files) offline
Sysprep	System Preparation Tool — generalizes a Windows installation (removes unique identifiers like the SID) so it can be captured as a master image for deployment
oscdimg.exe	An ADK command-line tool that creates ISO images from a directory structure; required when you need to produce a bootable, self-contained deployment ISO
WinPE	Windows Preinstallation Environment — a minimal Windows environment used to boot a machine for deployment, recovery, or diagnostics before the full OS is installed

Lab exercises (SSW-Lab)

VM	Task
LAB-DC01	Verify domain `ssw.lab` — run `Get-ADDomain` in PowerShell
LAB-MGMT01	Install Windows ADK + Deployment Tools via `Build-UnattendedIsos.ps1`
LAB-W11-01	Verify Windows 11 version: `winver` → note the build number
LAB-W11-01	Run in-place upgrade simulation: analyse `Get-WindowsUpdateLog`
LAB-MGMT01	Create an answer file using Windows System Image Manager (SIM)

Lab commands

```
# Check current Windows build number
(Get-ItemProperty 'HKLM:\SOFTWARE\Microsoft\Windows
NT\CurrentVersion').CurrentBuild

# Mount a WIM image with DISM for offline servicing
dism /Mount-Image /ImageFile:"C:\Images\install.wim" /Index:1
/MountDir:"C:\Mount"

# Unmount and commit changes
dism /Unmount-Image /MountDir:"C:\Mount" /Commit

# Generalise the reference machine before capture
C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown

# Create a bootable ISO from a directory (Windows ADK)
oscdimg.exe -n -m -bC:\WinPE\boot\etfsboot.com C:\WinPE_x64 C:\Output\WinPE.iso
```

Knowledge check

1. What is the difference between a *wipe-and-load* and an *in-place upgrade*?
2. What is the minimum build Windows 11 requires for Intune enrollment?
3. What does `oscdimg.exe` do and why is it needed for unattended deployments?
4. When do you use DISM versus sysprep?

▶ Answers

Week 2 — Intune enrollment and device management

> **Exam domain:** Prepare infrastructure for devices · **Weight:** 25–30%

> **Real-world scenario:** A Sogeti consultant is onboarding a new client — a 200-person professional services company that has no on-premises domain infrastructure. The client wants all Windows 11 laptops managed via Intune with BitLocker enforced on every device. New employees receive laptops directly from the hardware vendor and should be able to self-provision without calling the helpdesk. The consultant must choose the right enrollment method, configure a BitLocker enforcement profile, and ensure the compliance dashboard is operational before go-live.

Learning Objectives

- [ ] Explain the difference between MDM enrollment and Hybrid Azure AD Join and identify when each approach is appropriate
- [ ] List and compare the Intune enrollment methods available for Windows devices (manual, Autopilot, bulk/PPKG, GPO auto-enrollment)
- [ ] Create a Configuration Profile in Intune and assign it to a device group
- [ ] Enforce BitLocker via an Intune configuration profile and verify the encryption status using `manage-bde`
- [ ] Create a device compliance policy that checks OS version, and understand how compliance status feeds into Conditional Access
- [ ] Interpret the *Compliant*, *Not compliant*, *In grace period*, and *Not evaluated* device statuses in the Intune portal

MS Learn modules

- Enroll devices in Microsoft Intune
- Manage device profiles with Intune
- Monitor devices with Intune

Key Concepts

Term	Description
MDM enrollment	Full device management via Mobile Device Management protocol; Intune receives authority over the device and can push policies, apps, and remote actions
Hybrid Azure AD Join	The device is joined to an on-premises Active Directory domain AND registered in Entra ID; enrollment in Intune is typically triggered via Group Policy auto-enrollment
Azure AD Join	The device is joined exclusively to Entra ID with no on-premises AD dependency; the primary join type for cloud-first organisations and Autopilot deployments
Configuration profile	An Intune policy object that configures a specific set of device settings (Wi-Fi, BitLocker, VPN, restrictions, etc.) and is assigned to users or device groups
Compliance policy	A set of rules defining what conditions a device must meet to be considered *compliant* (minimum OS version, BitLocker enabled, Defender running, etc.)
Enrollment Status Page (ESP)	A screen displayed during Autopilot provisioning that blocks the user from accessing the desktop until all assigned apps and profiles have been installed
manage-bde	Built-in Windows command-line tool for querying and managing BitLocker encryption status on a drive; `manage-bde -status C:` shows encryption state
MDM Diagnostic Tool	`MdmDiagnosticsTool.exe -out <path>` collects an enrollment diagnostics bundle from a Windows device, useful for troubleshooting enrollment failures

Lab exercises (SSW-Lab)

VM	Task
LAB-W11-01	Enroll device in Intune via **Settings** → **Accounts** → **Work or school**
LAB-W11-02	Enroll second device — observe the difference in the Intune portal
LAB-MGMT01	Open Intune portal (intune.microsoft.com) → verify both devices under **Devices** → **All devices**
LAB-MGMT01	Create a *Configuration profile*: BitLocker enforced on W11-01
LAB-W11-01	Verify BitLocker status: `manage-bde -status`
LAB-MGMT01	Review **Device compliance** — create a compliance policy (minimum OS version)

Lab commands

```
# Check BitLocker encryption status on C: drive
manage-bde -status C:

# Force an immediate Intune policy sync from the device
Start-Process "ms-device-enrollment:// enroll"
# Alternatively, trigger sync via Scheduled Task:
Start-ScheduledTask -TaskPath "\Microsoft\Windows\EnterpriseMgmt\" -TaskName
"Schedule to run OMADMClient by client"

# Collect MDM diagnostic bundle for enrollment troubleshooting
MdmDiagnosticsTool.exe -out C:\Temp\MdmDiag

# Check Intune enrollment status in registry
Get-ItemProperty "HKLM:\SOFTWARE\Microsoft\Enrollments\*" | Select PSChildName,
EnrollmentType, UPN
```

Knowledge check

1. What is the difference between MDM enrollment and Hybrid Azure AD Join?
2. What enrollment methods exist in Intune and when do you use each?
3. What does a *Compliant* versus *Not compliant* status mean in Intune?
4. How does the *Enrollment Status Page* work in Autopilot?

▶ Answers

Week 3 — Compliance, Conditional Access and identity

> **Exam domain:** Prepare infrastructure for devices · **Weight:** 25–30%

> **Real-world scenario:** A Sogeti consultant is engaged by a manufacturing company after a targeted phishing attack compromised several employee accounts. The CISO mandates that all cloud application access must require MFA when users are off-site, and that only Intune-enrolled, compliant devices can access Microsoft 365 services. The consultant must configure Conditional Access policies, set up Named Locations for the corporate network, deploy Windows LAPS to prevent lateral movement via shared local admin passwords, and ensure the hybrid identity sync is working correctly via Azure AD Connect.

Learning Objectives

- [ ] Describe the three categories of signals that Conditional Access evaluates (user/group, device condition, location/risk) and explain how they combine into an access decision
- [ ] Distinguish between *Block access* and *Grant access with controls* in a Conditional Access policy, and when to use each
- [ ] Compare Azure AD Connect Sync and Microsoft Entra Cloud Sync, including their supported topologies and limitations
- [ ] Configure Windows LAPS via Intune (backup destination, password age, rotation) and retrieve a managed local admin password
- [ ] Explain the difference between Windows LAPS (modern, cloud-integrated) and legacy Microsoft LAPS (AD-only, requires separate agent)
- [ ] Use Named Locations in Conditional Access to define trusted IP ranges and enforce location-based access controls

MS Learn modules

- Configure device compliance policies
- Configure Conditional Access
- Manage user and device identities

Key Concepts

Term	Description
Conditional Access	An Entra ID policy engine that evaluates signals (user identity, device compliance, location, sign-in risk) and grants, blocks, or grants with controls (e.g., MFA required)
Named Locations	Defined IP address ranges or countries in Entra ID that CA policies can use as a condition — e.g., "require MFA when signing in from outside the corporate IP range"

Term	Description
Azure AD Connect Sync	On-premises sync agent that synchronises users, groups, and devices from AD DS to Entra ID; supports complex, multi-forest topologies; requires Windows Server on-premises
Entra Cloud Sync	Lightweight cloud-based sync agent that supports simpler hybrid identity scenarios; managed entirely from the cloud with no on-premises infrastructure beyond the lightweight agent
Windows LAPS	Local Administrator Password Solution built into Windows 11 22H2+ and Windows Server 2022; automatically rotates the local admin password and backs it up to Entra ID or AD DS
Legacy LAPS	The original Microsoft LAPS solution (client-side extension); backs up passwords only to on-premises AD, requires a separate MSI agent install, and is configured via Group Policy
Primary Refresh Token (PRT)	A long-lived SSO token issued to Entra ID-joined or registered devices after user authentication; used to silently acquire access tokens for cloud resources without re-prompting for credentials
Report-only mode	A CA policy state where the policy evaluates sign-ins and logs what it would have done, without actually enforcing a block or grant — safe way to test CA policy impact before enabling it

Lab exercises (SSW-Lab)

VM	Task
LAB-DC01	Create test users in AD: `New-ADUser -Name "TestUser01" ...`
LAB-DC01	Sync AD to Entra ID via Azure AD Connect (install on DC01)
LAB-MGMT01	Configure a Conditional Access policy: MFA required outside corporate network
LAB-W11-01	Test the CA policy: sign in as TestUser01, verify MFA prompt
LAB-MGMT01	Create a compliance policy: requires Defender, BitLocker, and min. W11 22H2
LAB-W11-02	Demonstrate non-compliant device → verify block in CA policy
LAB-MGMT01	Enable **Windows LAPS** in Intune: Endpoint security → Account protection → LAPS policy
LAB-W11-01	Verify LAPS: retrieve the rotated local admin password via Intune portal

Lab commands

```
# Create a test user in Active Directory
New-ADUser -Name "TestUser01" -SamAccountName "testuser01" -UserPrincipalName
"testuser01@ssw.lab" `
  -AccountPassword (ConvertTo-SecureString "P@ssw0rd!" -AsPlainText -Force) -
Enabled $true

# Force an Azure AD Connect delta sync
Start-ADSyncSyncCycle -PolicyType Delta

# Retrieve the LAPS-managed local admin password via Microsoft Graph (requires Az
or Graph module)
Get-MgDeviceLocalCredentials -DeviceId "<device-object-id>" -IncludeSecrets

# Check Windows LAPS status on the local device
Get-LapsAADPassword -DeviceId (Get-ItemProperty "HKLM:\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\Winlogon").DefaultDomainName
```

Knowledge check

1. What signals does Conditional Access use to make an access decision?
2. What is the difference between *Block* and *Grant with controls* in CA?
3. How does Azure AD Connect Sync compare to Cloud Sync?
4. What does the *Named Locations* setting do in CA?
5. What is *Windows LAPS* and how does it differ from the legacy LAPS solution?

▶ Answers

Week 4 — Application management with Intune

> **Exam domain:** Manage applications · **Weight:** 15–20%

> **Real-world scenario:** A Sogeti consultant is supporting a retail chain that manages 600 Windows 11 endpoints entirely via Intune. The IT team needs to deploy a legacy ERP application (complex EXE installer with custom switches) silently to all corporate devices, make the Adobe Acrobat Reader available optionally in the Company Portal, and ensure that employees using personal iPhones for work email cannot copy corporate data to their personal apps. The consultant must package the Win32 app, configure app assignment intents, and set up App Protection Policies for mobile BYOD.

Learning Objectives

- ☐ Package a Win32 application using the Intune Win32 Content Prep Tool (`IntuneWinAppUtil.exe`) and upload it to Intune with a correct detection rule
- ☐ Distinguish between *Required*, *Available*, and *Uninstall* app assignment intents and explain when each is appropriate
- ☐ Describe the role and location of `IntuneManagementExtension.log` and use it to diagnose Win32 app deployment failures
- ☐ Deploy Microsoft 365 Apps via the built-in Intune app type and understand the relationship between Intune, ODT, and OCT
- ☐ Configure an App Protection Policy (MAM) and explain how MAM-without-enrollment protects data on unmanaged BYOD devices
- ☐ Explain when to use Win32 app packaging versus the Microsoft Store (new) or the Enterprise App Catalog

MS Learn modules

- Deploy and update applications with Intune
- Manage Win32 apps with Intune
- Configure Microsoft 365 Apps deployment

Key Concepts

Term	Description
IntuneWinAppUtil.exe	Free Microsoft command-line tool that wraps an application installer and its dependencies into a `.intunewin` package for upload to Intune
Win32 app	The most flexible Intune app type — supports any installer format (EXE, MSI, MSIX) packaged as `.intunewin`; supports custom install/uninstall commands and detection rules
Detection rule	Logic that Intune uses to determine if a Win32 app is already installed; can be based on file path existence, a registry key/value, or MSI product code
Required assignment	The app is forcibly installed (or uninstalled) on all devices/users in the assigned group without user interaction
Available assignment	The app appears in Company Portal and the user can choose to install it themselves; not pushed automatically
IntuneManagementExtension.log	The primary log file for Win32 app deployments on Windows devices; located at `C:\ProgramData\Microsoft\IntuneManagementExtension\Logs\`; contains install command output, detection results, and error codes
App Protection Policy (MAM)	A set of rules that protect corporate data within managed apps (e.g., Outlook, Teams, Edge) — encrypts app data, restricts copy/paste to unmanaged apps, and can wipe corporate data selectively; works with or without MDM enrollment

Term	Description
MAM-WE	Mobile Application Management Without Enrollment — applies App Protection Policies to corporate apps on BYOD devices that are NOT enrolled in Intune; only the app-level data is managed, not the device
ODT / OCT	Office Deployment Tool (command-line installer for Microsoft 365 Apps) and Office Customization Tool (web GUI at config.office.com for generating the XML configuration); used together for advanced M365 Apps deployments

Lab exercises (SSW-Lab)

VM	Task
LAB-MGMT01	Package a `.exe` app as `.intunewin` using the **Intune Win32 Content Prep Tool**
LAB-MGMT01	Upload the Win32 app to Intune → assign to W11-01 (Required)
LAB-W11-01	Verify installation via **Company Portal** or event log (`IntuneManagementExtension`)
LAB-MGMT01	Create a Microsoft 365 Apps deployment in Intune (Office suite)
LAB-W11-02	Verify Office installation after sync (`imdssync` or wait for Intune check-in)
LAB-MGMT01	Configure an *App protection policy* (MAM) for Microsoft Edge

Lab commands

```
# Package a Win32 app installer into .intunewin format
.\IntuneWinAppUtil.exe -c "C:\AppSource\MyApp" -s "setup.exe" -o
"C:\IntunePackages"

# Tail the IntuneManagementExtension log in real time for troubleshooting
Get-Content
"C:\ProgramData\Microsoft\IntuneManagementExtension\Logs\IntuneManagementExtension.
-Wait -Tail 50

# Check if a specific app is detected (simulate detection rule - file path)
Test-Path "C:\Program Files\MyApp\myapp.exe"

# Trigger IME app re-evaluation (restart the service)
Restart-Service -Name IntuneManagementExtension
```

Knowledge check

1. What is the difference between a *Required* and *Available* app assignment?
2. When do you use Win32 app packaging versus Microsoft Store for Business?

3. What does `IntuneManagementExtension.log` contain and where is it located?
4. What are the benefits of MAM without MDM enrollment?

▶ Answers

Week 5 — Windows Autopilot

> **Exam domain:** Prepare infrastructure for devices · **Weight:** 25–30%

> **Real-world scenario:** A Sogeti consultant is setting up a zero-touch deployment process for a logistics company that receives 50 new laptops per month from a hardware reseller. The company wants new employees to unbox their laptop, connect to the internet, and have a fully configured, corporate-ready Windows 11 device within 30 minutes — without any IT involvement. Additionally, the company has shared kiosk terminals in warehouses that must auto-configure without any user sign-in. The consultant must configure Autopilot for both User-Driven and Self-Deploying scenarios, set up the Enrollment Status Page, and register the hardware hashes via the reseller portal.

Learning Objectives

- ☐ Collect a device hardware hash using `Get-WindowsAutoPilotInfo` and import it into the Intune portal
- ☐ Configure an Autopilot deployment profile for User-Driven mode with Entra ID join and walk through the OOBE end-to-end
- ☐ Differentiate between User-Driven, Self-Deploying, Pre-Provisioning (White Glove), and Autopilot for Existing Devices scenarios
- ☐ Configure the Enrollment Status Page and explain how it controls app installation sequencing during provisioning
- ☐ Perform an Autopilot Reset on a registered device and explain when it is used versus a full wipe
- ☐ Explain the positioning of Windows 365 versus Azure Virtual Desktop and identify the correct scenario for each
- ☐ Run a basic KQL device query in Intune using the Device query feature and interpret the results

MS Learn modules

- Configure Windows Autopilot
- Autopilot deployment scenarios
- Troubleshoot Windows Autopilot

Key Concepts

Term	Description
Hardware hash	A cryptographic fingerprint of the device's hardware components (motherboard, NIC, etc.) used to uniquely identify the device for Autopilot registration; collected via `Get-WindowsAutoPilotInfo`
User-Driven mode	Autopilot scenario where a user authenticates with their Entra ID account during OOBE, and the device is provisioned with their profile and assigned apps; most common enterprise deployment mode
Self-Deploying mode	Autopilot scenario with no user interaction during OOBE; the device authenticates using TPM 2.0 device attestation; designed for kiosks, shared devices, or digital signage
Pre-Provisioning (White Glove)	A two-phase Autopilot mode where IT or a reseller completes the device-targeted provisioning phase (apps, profiles) in advance, then the user completes the user-targeted phase when they first sign in
Group Tag	A string label attached to an Autopilot device that can be used in dynamic Entra ID group membership rules (`(device.devicePhysicalIds -any _ -eq "[OrderID]:TAGVALUE")`) to automatically assign profiles
Autopilot Reset	A Windows action that reinstalls Windows while keeping the device registered in Autopilot and Entra ID; preserves corporate settings but wipes user data; useful for reassigning a device to a new user
Windows 365 Cloud PC	Desktop-as-a-Service offering from Microsoft; each user gets a persistent, personal virtual desktop with a fixed monthly per-user price; managed via Intune like a physical device
Azure Virtual Desktop (AVD)	Microsoft's Azure-based VDI platform; supports multi-session Windows (multiple users on a single VM host), pay-per-use pricing, and pooled/non-persistent desktops; more complex and flexible than Windows 365
KQL Device Query	Kusto Query Language queries run per-device in real-time via Intune Advanced Analytics; retrieves current hardware, software, user, and disk information directly from the device without waiting for inventory sync

Lab exercises (SSW-Lab)

VM	Task
LAB-W11-AUTOPILOT	Gather hardware hash: `Get-WindowsAutoPilotInfo -OutputFile hash.csv`
LAB-MGMT01	Upload hash to Intune: **Devices** → **Windows** → **Enrollment** → **Windows Autopilot devices**
LAB-MGMT01	Create an Autopilot deployment profile: *User-driven, Microsoft Entra join*
LAB-W11-AUTOPILOT	Reset the VM (Settings → System → Recovery → Reset this PC)

VM	Task
LAB-W11-AUTOPILOT	Walk through the Out-of-Box Experience (OOBE) → verify automatic enrollment
LAB-MGMT01	Analyse Autopilot events in **Event Viewer** → **Applications and Services** → **Microsoft** → **Windows** → **Autopilot**
LAB-MGMT01	Explore *Windows 365* in Intune: review Cloud PC provisioning policies
LAB-MGMT01	Run a *device query* using KQL: **Devices** → **select device** → **Device query**

Lab commands

```
# Collect hardware hash and export to CSV (requires WindowsAutoPilotIntune
module)
Install-Script -Name Get-WindowsAutoPilotInfo
Get-WindowsAutoPilotInfo -OutputFile C:\Temp\AutopilotHash.csv

# Upload hardware hash to Intune directly (requires MS Graph /
WindowsAutoPilotIntune module)
Install-Module -Name WindowsAutoPilotIntune
Connect-MgGraph -Scopes "DeviceManagementServiceConfig.ReadWrite.All"
Import-AutoPilotCSV -csvFile C:\Temp\AutopilotHash.csv

# KQL device query examples in Intune Device Query
# Run these in: Intune → Devices → [select device] → Device query
# InstalledApplications | project ApplicationName, ApplicationVersion | order by
ApplicationName asc
# SystemInfo | project DeviceName, Manufacturer, Model, OSVersion, TotalMemory
```

Knowledge check

1. What is the difference between *User-driven* and *Self-deploying* Autopilot mode?
2. What is the *Enrollment Status Page* used for and how do you configure it?
3. How do you reset an Autopilot profile assignment if a device is already registered?
4. What is *Windows Autopilot Reset* and when do you use it?
5. What is *Windows 365* and how does it differ from Azure Virtual Desktop?
6. How do you run a KQL device query in Intune and what data can you retrieve?

▶ Answers

Week 6 — Security, updates, Intune Suite and monitoring

> **Exam domain:** Protect devices · **Weight:** 15–20%

> **Real-world scenario:** A Sogeti consultant is engaged by a healthcare organisation following a ransomware incident that encrypted several file servers. The CISO requires immediate action: enforce Windows Update patching within 7 days of release, onboard all endpoints to Microsoft Defender for Endpoint, remove local administrator rights from all standard users while still allowing certain maintenance tools to run with elevation, and implement a proactive monitoring strategy that alerts IT before devices become a problem. The consultant must configure update rings, Defender onboarding, Endpoint Privilege Management, and Endpoint Analytics.

Learning Objectives

- ☐ Configure a Windows Update for Business update ring in Intune and differentiate between quality update and feature update deferral periods
- ☐ Distinguish between an *Update ring* (WUfB) and a *Feature update policy* and explain when to use each
- ☐ Enable Microsoft Defender for Endpoint onboarding via Intune and verify sensor status using the Defender portal and the `sc query sense` command
- ☐ Explain how co-management works between Intune and Configuration Manager and identify which workloads to shift first
- ☐ Describe the Intune Suite add-on capabilities (EPM, Remote Help, Advanced Analytics, Enterprise App Catalog, Cloud PKI, Tunnel for MAM) and the problem each solves
- ☐ Use Intune remote actions (wipe, retire, sync, rotate LAPS password) and explain the difference between *Wipe* and *Retire*

MS Learn modules

- Manage endpoint security with Intune
- Manage Windows updates with Intune
- Monitor and troubleshoot devices
- Intune Suite add-on capabilities

Key Concepts

Term	Description
Update ring	A Windows Update for Business policy in Intune that defines deferral periods for quality updates and feature updates, active hours, restart behaviour, and deadline grace periods
Quality update	Monthly cumulative security and bug-fix updates released on Patch Tuesday; deferral is configurable up to 30 days

Term	Description
Feature update	Biannual Windows version upgrades (e.g., 22H2 → 23H2); deferral configurable up to 365 days; requires a separate *Feature update policy* in Intune for controlled targeting
Deferral period	Number of days after Microsoft releases an update before it is offered to devices in the ring; used to give time for testing before broad deployment
Endpoint analytics	Intune dashboard showing device health scores, startup performance, app reliability, and proactive remediations results across the fleet
Co-management	A configuration where a Windows device is simultaneously managed by Configuration Manager (SCCM) and Microsoft Intune; workloads (Compliance, Endpoint Protection, etc.) can be individually shifted from ConfigMgr to Intune
Wipe	A remote action that factory-resets a device by removing all data and reinstalling Windows; the device is removed from Entra ID and Intune; use when a device is lost or being decommissioned
Retire	A remote action that removes corporate data and configuration from the device while leaving personal data intact; the device is unenrolled from Intune; appropriate for BYOD or when an employee leaves
Endpoint Privilege Management (EPM)	An Intune Suite add-on that allows standard (non-admin) users to run specific, IT-approved processes with elevated privileges without making them local administrators
Microsoft Tunnel for MAM	An Intune Suite add-on VPN tunnel for MAM-managed apps on non-enrolled iOS/Android devices; only managed app traffic routes through the tunnel — the device itself does not need to be enrolled

Lab exercises (SSW-Lab)

VM	Task
LAB-MGMT01	Configure an *Update ring* in Intune: Semi-Annual Channel, 7-day deferral
LAB-W11-01	Check Windows Update status: `Get-WindowsUpdateLog`
LAB-MGMT01	Enable Microsoft Defender for Endpoint via Intune *Endpoint security* → *Antivirus*
LAB-W11-01	Run a Defender Quick Scan: `Start-MpScan -ScanType QuickScan`
LAB-W11-02	Simulate a detection with EICAR test file → analyse alert in Defender portal
LAB-MGMT01	View **Device diagnostics** in Intune portal → download diagnostics from W11-01
LAB-MGMT01	Explore *Endpoint Privilege Management* (EPM) in Intune Suite: create an elevation policy
LAB-MGMT01	Review *Advanced Analytics* in Intune: check anomaly detection dashboard
LAB-MGMT01	Explore **Enterprise App Catalog**: find a managed app and review its metadata

Lab commands

```
# Check Windows Defender service (SENSE = MDE sensor) status
sc query sense

# Trigger a Defender Quick Scan
Start-MpScan -ScanType QuickScan

# Check current Windows Update deferral settings in registry
Get-ItemProperty "HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate" |
Select DeferQualityUpdates, DeferQualityUpdatesPeriodInDays

# Check Defender threat definitions version
Get-MpComputerStatus | Select AntivirusSignatureVersion,
AntispywareSignatureLastUpdated, RealTimeProtectionEnabled

# Initiate a Defender signature update
Update-MpSignature
```

Knowledge check

1. What is the difference between an *Update ring* and a *Feature update policy*?
2. How does *Co-management* work between Intune and Configuration Manager?
3. What does the **Endpoint analytics** dashboard show in Intune?
4. How do you use *Remote actions* (wipe, retire, sync) in Intune?
5. What Intune Suite add-ons exist and what problem does each solve?
6. What is *Endpoint Privilege Management* and when do you use elevation policies?

▶ Answers

Week 7 — Exam preparation

Activities

- Review weak domains based on the official exam study guide
- Complete the **Microsoft Learn practice assessment** for MD-102: Practice assessment
- Revisit lab tasks you felt less confident about (week 2 Intune enrollment, week 5 Autopilot)
- Create a summary of all PowerShell commands used in the lab exercises
- Schedule your exam via Pearson VUE or Certiport

Exam focus areas

- **Prepare infrastructure (25–30%):** Entra join types, enrollment methods, compliance policies, Conditional Access, Windows LAPS

- **Manage & maintain devices (30–35%):** Autopilot deployment modes, configuration profiles, Windows 365 vs AVD, KQL device queries, Intune Suite add-ons (EPM, Remote Help, Tunnel for MAM)
- **Manage applications (15–20%):** Win32/LOB/Store/M365 Apps, app protection policies, ODT/OCT, Enterprise App Catalog
- **Protect devices (15–20%):** Security baselines, Defender for Endpoint onboarding, update rings vs feature update policies

Common exam traps — know these cold

Trap	The correct answer
Self-Deploying Autopilot requires a user to sign in	**No** — zero user interaction; device authenticates via TPM 2.0 attestation
MAM requires device enrollment	**No** — MAM-WE (without enrollment) applies App Protection Policies to apps on unmanaged BYOD devices
A non-compliant device is immediately blocked	**No** — a configurable grace period applies before the block takes effect
Security Baseline always overrides Configuration Profiles	**No** — conflicts between policies are reported as Errors; no automatic winner
BitLocker always prompts the user during encryption	**No** — Intune supports silent BitLocker encryption via TPM with no user interaction
Tenant Attach equals Co-management	**No** — Tenant Attach gives visibility of ConfigMgr devices in the Intune portal; co-management means shared workload authority
Autopilot Reset requires reimaging	**No** — Autopilot Reset re-provisions Windows while keeping the device's Autopilot registration and Entra ID object intact
The Enterprise App Catalog requires manual packaging	**No** — apps in the catalog are pre-packaged by Microsoft with detection rules and auto-updates included

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Exam Coverage Gaps and Must-Do Labs

This section identifies topics that are tested on the MD-102 exam but are not fully covered by the 7-week SSW-Lab schedule. Complete these before scheduling your exam.

Topics not fully covered by the weekly labs

1. **Cross-platform enrollment and profiles** — The lab schedule focuses on Windows. The exam tests knowledge of Android Enterprise enrollment modes (Fully Managed, Dedicated, Corporate-Owned Work Profile, Personally-Owned Work Profile), iOS/iPadOS ADE (Automated Device Enrollment via Apple Business Manager), and macOS configuration profiles. Study these conceptually via MS Learn even if you cannot test them in the SSW-Lab environment.

2. **Bulk enrollment via Provisioning Package (PPKG)** — Creating a PPKG with Windows Configuration Designer and applying it during OOBE (via USB) or post-setup (`Add-ProvisioningPackage`). Know when PPKG is the right tool: offline deployments, no Intune license, devices without TPM 2.0.

3. **Delivery Optimization** — Intune policy for peer-to-peer Windows Update content sharing within a subnet, reducing internet bandwidth usage. Know the Download mode options (HTTP only, LAN peering, Internet peering, Simple, Bypass) and when to configure each.

4. **Security Baselines** — Deploying the Windows Security Baseline, Microsoft Defender for Endpoint Baseline, and Microsoft Edge Baseline in Intune (Endpoint security → Security baselines). Understand that baselines are opinionated bundles of recommended settings, not individual profiles, and that conflicting settings between a baseline and a configuration profile are reported as errors without a defined winner.

5. **Attack Surface Reduction (ASR) rules** — Common rules tested on the exam: "Block Office apps from creating executable content", "Block credential stealing from LSASS", "Block JavaScript/VBScript from launching downloaded executables". Know the three modes: Audit (log only), Block (enforce), and Warn (block with user override).

6. **Intune Suite components not tested in the lab** — Remote Help (setup, RBAC roles, session audit logs), Cloud PKI (replaces ADCS + NDES + Intune Certificate Connector), and Microsoft Tunnel for MAM (per-app VPN for non-enrolled mobile devices). Understand the architectural difference between each add-on and the problem it solves.

7. **Co-management workload migration** — Know the co-management slider, the supported workloads, the concept of the Pilot collection, and the recommended sequence for shifting workloads from ConfigMgr to Intune.

Must-do labs before sitting the exam

1. Create at least one **Android Enterprise enrollment profile** (Fully Managed) in your Intune tenant. Document which settings are exam-relevant (token expiry, QR code provisioning, zero-touch enrollment requirement).

2. Create at least one **iOS/iPadOS configuration profile** with Device restrictions applied via a filter. Note which settings are available that differ from Windows profiles.

3. Configure an **Update ring** (quality deferral 7 days, feature deferral 30 days) and a **Feature update policy** (pin to a specific Windows 11 version) side by side in your Intune tenant. Document the difference in what each controls.

4. Run at least five KQL **device queries** in Intune using the `InstalledApplications`, `SystemInfo`, `LocalUsers`, and `LogicalDrive` tables. Include the query output in your study notes.

5. Perform five complete end-to-end scenarios and document the steps: (a) Enroll a device, (b) create and assign a compliance policy, (c) configure a Conditional Access policy that blocks non-compliant devices, (d) deploy a Required Win32 app, (e) trigger a remote action (Sync, then Retire).

6. Create and test a **Proactive Remediation** (detection script + remediation script pair). Verify that Intune reports the detection and remediation status correctly in the portal.

Exit criteria — do not schedule the exam until you can answer all of these

1. You can explain all four exam domains with concrete examples drawn from your own lab work (not just from reading).
2. You have performed at least two hands-on labs per exam domain.
3. You score consistently above 75% on the Microsoft Learn practice assessment and can explain — not just identify — why each incorrect answer is wrong.
4. You can draw the Autopilot provisioning flow (OOBE → ESP phases → desktop) from memory.
5. You can explain the difference between these six pairs without hesitation: (Wipe vs Retire) · (Update ring vs Feature update policy) · (Windows LAPS vs Legacy LAPS) · (Co-management vs Tenant Attach) · (Windows 365 vs AVD) · (MAM-WE vs MDM enrollment).